

Helix OTA — Continuation

Revision: 5 Last modified: 2026-06-22T19:30:00Z

1. Current state

Field	Value
HEAD	e1abfe6c
Phase	Stable / release-readiness — control plane proven on real RK3588 hardware; native A/B fidelity now routed through the new Cuttlefish containerized path (integration-pending on Linux+KVM)
Terminal goal	Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets

Latest session (2026-06-22, evening) — Cuttlefish Tier-2 container path + real RK3588 control-plane validation

Three new real capabilities landed (honest §11.4.6 — boundaries stated, not overclaimed):

- Cuttlefish Tier-2 containerized path (F112)** — new pkg/cuttlefish cvd-lifecycle wrapper in the containers submodule (cuttlefish.go / accel.go / cleanup.go / health.go / types.go / entrypt.sh / Containerfile). go test -race ./pkg/cuttlefish/ = **30 PASS
 - 1 honest topology SKIP** (no Linux+KVM on this macOS host). Rootless cannot host Cuttlefish, so a narrowest-scope **rootful-privileged documented exception** is recorded — docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md (§11.4.161 documented exception via §11.4.112; image build + artifact fetch stay rootless, only launch_cvd is privileged). **HONEST BOUNDARY: the container path is BUILT + unit-tested — it is NOT yet a real-A/B PASS.** The real end-to-end Android update_engine/AVB/dm-verity A/B run is **integration-pending** on nezha Linux+KVM (assets staging; privileged launch_cvd operator-gated). Native A/B fidelity is this path’s job once provisioned.
- Real RK3588 hardware control-plane validation (F113)** — evidence docs/qa/20260622-rk3588-controlplane/REPORT.md. Server cross-built linux/

amd64, run **rootless** on nezha (uid 1000). **Device B** (Ethernet, serial 1acdceab90248933) **PASS**: board-originated GET /healthz 200, GET /api/v1/client/update 204 (no active deployment — correct), POST /api/v1/client/telemetry 202, and sink-side GET /devices/by-hardware/1acdceab90248933 → update_state=success (the board’s own telemetry mutated server state). **Device A** (Wi-Fi, 19bbb528a1dbbc4d) honest topology **SKIP** — VPN tun1 full-tunnel / Wi-Fi AP isolation (busybox nc to both :18080 and :22 time out → blocked network path, captured root cause, NOT a Helix defect). **HONEST BOUNDARY: both boards are NON-A/B** (single-slot, no update_engine) → this validates the control plane on real hardware, NOT native A/B apply. **ZERO device state changes** (§11.4.122/§11.4.133).

3. **Distribution repoint** — container distribution targets now thinker.local (live) + amber.local (SSH-key-pending); nezha.local is read/import-only.

Operator action items: (1) **amber.local** — install the SSH key (ssh-copy-id to amber) to bring it live as a distribution target; (2) **Cuttlefish on nezha** — the privileged launch_cvd needs operator sudo (+ reboot + ~30 GB fetch_cvd) to unblock the real Android A/B run (F112 integration-pending); (3) **assets staging in progress** for the Cuttlefish Tier-2 run; (4) the **physical RK3588 boards are NON-A/B** (single-slot) — real on-device A/B fidelity is the Cuttlefish path’s job, not these boards (control-plane validation only on them).

Prior session (2026-06-22, daytime) — “do everything workable” sweep

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, -race 0); **gofmt fixed** (13 files); pre-build gates **PASS**; docs_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence docs/qa/20260622T07*. Fixed a real §11.4.115 RED-mode polarity isolation bug in ab_rauc_verity.sh (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — docs/qa/20260622-server-recordings-regen/; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep**: propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into commit_all.sh; constitution pin → 09d8940 (adds docs/semgrep/TOKEN_SETUP.md); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (ota-update-engine-bridge, ota-android-agent); GitFlic “blocker” disproven; §11.4.30 transient qa-results/ untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch_cvd).

Operator action items: (1) Semgrep SEMGREP_APP_TOKEN — follow constitution/docs/semgrep/TOKEN_SETUP.md (semgrep login) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

Known robustness gap (tracked, NOT yet fixed — §11.4.6/§11.4.123): scripts/boot_android_emulator.sh — the remote qemu/AVD on nezha stays tied to the launching SSH session and exits gracefully if that session ends mid-boot-wait (the nohup does not fully detach the remote process; the final SSH-tunnel/attestation step isn’t reached on interrupt). Boot itself is **PROVEN** (emulator-5554, API 36, boot_completed=1, evidence qa-results/20260622T071848Z-nezha-android-ab/). Fix candidate: wrap the remote launch in setsid nohup ... </dev/

null >log 2>&1 & (or a systemd-run --user --scope on nezha) for true detachment — deferred because on-target persistence verification (re-boot + interrupt test, leaves remote state) is operator-attended; a blind change to the working boot path is forbidden per §11.4.1 without rock-solid proof.

Active items

ID	Title	Status	Type
OTA-003	Emulator Tier-2 — real Android A/ B (update_engine/ AVB/dm-verity auto-rollback)	In testing	Task
OTA-004	Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U- Boot slot-switch, dm-verity on real partitions	Operator-blocked	Task

Recently closed items

ID	Title	Status
OTA-021	HelixTrack bidirectional sync verification	Completed
OTA-020	Database migration test	Fixed
OTA-019	Build resource stats tracker	Completed
OTA-018	ApplyPort CLI + slot manager + Ed25519 verifier	Implemented
OTA-017	U-Boot corrupt-slot auto- rollback via bootcount	Implemented
OTA-016	RAUC dd-apply to inactive slot with dm- verity	Implemented
OTA-015	A/B slot switch via U- Boot BOOT_ORDER	Implemented

Full issue details: docs/Issues.md · docs/Fixed.md · docs/
Issues_Summary.md · docs/Fixed_Summary.md.

2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok internal/api
ok internal/config
ok internal/device
ok internal/deviceemu
ok internal/fabric
ok internal/health
ok internal/rollout
ok internal/store
ok internal/transport
ok tests/chaos
ok tests/stress
```

No-test-file packages (build-only): cmd/applyport, cmd/ota-device-emu, cmd/ota-server, tools/loadtest.

3. Emulator status

Property	Value
Host	nezha.local — Linux x86_64, 62 GB RAM, KVM
Image	Android API 36 (Android 16) — CZ_API36_Phone
Transport	ADB over SSH tunnel (reachable at emulator-5554 from nezha)
ADB local	No Android devices connected locally (emulator is remote via SSH tunnel)
LD_LIBRARY_PATH	/home/milosvasic/.local/lib
Cuttlefish Tier-2	Pending AOSP guest images

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ_API36_Phone image provides the standard Android 16 AOSP stack on an x86_64 KVM host.

4. What's running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
 - **Background agents:** None currently dispatched
 - **Recording directory:** \$HOME/Downloads per §11.4.158(D) (default; no project-level override)
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5. Next actions (priority-ordered)

1. **OTA-003 — Emulator Tier-2 inline testing.** Verify the Android emulator on nezha.local can register with the control plane, receive an OTA payload, apply it

- via `update_engine`, A/B slot-switch, and auto-rollback on corruption. Drive end-to-end through the real user-equivalent path per §11.4.143.
2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity).
 3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
 4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.
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6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
 - **No-force-push (§11.4.113):** `git push --force`, `--force-with-lease`, `+<ref>` are STRICTLY FORBIDDEN. Always merge onto latest main and push fast-forward-only.
 - **Commit via `commit_all.sh`:** All changes use the project's canonical commit wrapper. Direct `git add/git commit/git push` are never used.
 - **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.
 - **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
 - **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
 - **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via `vasic-digital/containers` submodule.
 - **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.
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7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 11, 2026-06-21). Summary companion: `docs/features/Status_Summary.md`.

8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (docs/CONTINUATION.md) and docs/Issues.md for the full active-item context. The single highest-priority next action is **Emulator Tier-2 inline testing on nezha.local** (OTA-003) — drive the Android emulator through the full OTA lifecycle against the control plane.